LOCATION AND DEPTH OF COUNCIL DRAINAGE POINT ASSUMED.
BUILDER/CONTRACTOR MUST CONFIRM THE EXACT LOCATION, DEPTH AND INVERT LEVEL OF THE EXISTING COUNCIL DRAINAGE POINT (LPOD) ON-SITE PRIOR TO COMMENCING DRAINAGE WORKS. ANY DISCREPANCIES MUST BE REFERRED BACK TO DESIGN ENGINEER TO AMEND AND RESUBMIT TO COUNCIL FOR APPROVAL. FEES APPLY.

NOTE:
DOWNPIPES TO CHARGE WATER TANKS. DIAGRAMMATIC LAYOUT PLAN SHOWING CONNECTING DOWNPIPES TO THE WATER TANKS. BUILDER TO ENSURE THAT THE DESIGN INTENT IS CARRIED AS SHOWN AND DOWN PIPES ARE LOCATED AS PER ARCHITECTURAL DRAWINGS/ROOF DESIGN. NOTE THAT THE DOWNPIPES WILL BE CONSTANTLY CHARGED TO A MAXIMUM HEIGHT OF THE WATER TANKS OVERFLOW PIPE LEVELS. ALL DOWNPIPES JOINTS HAVE TO BE SOLVENT GLUED AND WATER TIGHT IN ACCORDANCE WITH AUSTRALIAN PLUMBING CODES SPECIFICATIONS. PERFORMANCE SOLUTION FOR CHARGED PIPES IS NOT PART OF THIS DESIGN.

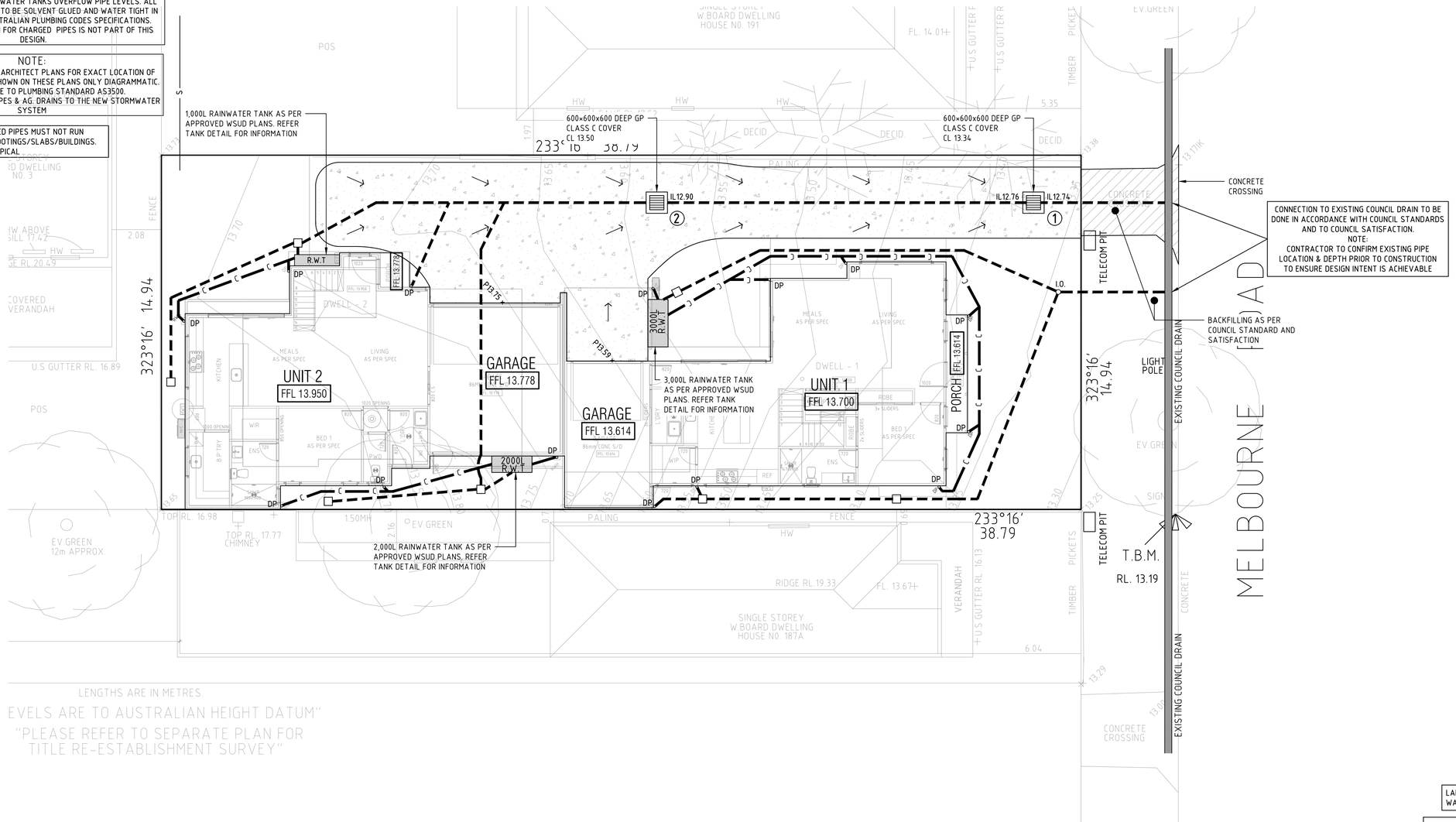
NOTE:
CONTRACTOR TO FOLLOW ARCHITECT PLANS FOR EXACT LOCATION OF DOWNPIPES. DOWNPIPES SHOWN ON THESE PLANS ONLY DIAGRAMMATIC. CONNECTION TO BE TO PLUMBING STANDARD AS3500.
CONNECT ALL NEW DOWNPIPES & AG DRAINS TO THE NEW STORMWATER SYSTEM

IN ANY CASE CHARGED PIPES MUST NOT RUN UNDER/THROUGH THE FOOTINGS/SLABS/BUILDINGS. TYPICAL

PROJECT ADDRESS:

NO 189 MELBOURNE ROAD WILLIAMSTOWN
SCALE 1:100

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LENGTHS ARE IN METRES.
LEVELS ARE TO AUSTRALIAN HEIGHT DATUM"
"PLEASE REFER TO SEPARATE PLAN FOR TITLE RE-ESTABLISHMENT SURVEY"

LEGEND

- PAVEMENT LEVEL
- IL 0.000 — INVERT LEVEL
- STORMWATER DRAIN U.N.O. ⌀150 UPVC @ 1:100 MIN (MIN SN8)
- 100⌀ UPVC @ 1:100 MIN (MIN SN10)
- SUSPENDED 100⌀ UPVC @ 1:100 MIN (MIN SN10)
- C — C — CHARGED/SEALED 100⌀ UPVC (MIN SN10)
- GRATED PIT (GP)
- JUNCTION PIT (JP)
- 150 — GRATED TRENCH (STORMWATER) NUMBER DENOTES WIDTH CLASS C COVER
- DP — DOWNPIPE
- DIRECTION OF SURFACE RUNOFF
- Ag — AGI DRAIN 90⌀
- SITE BOUNDARY — PROPERTY BOUNDARY
- B1 — BARRIER KERB - TYPE B1
- +NS.73.00 — SITE LEVELS (BY OTHERS)
- ×FS.73.00 — FINISH SURFACE LEVELS (BY OTHERS)
- R.W.T — RAIN WATER TANK
- SUMP POINT
- 350x350 GRATED PIT
- I.O. — INSPECTION OPENING

PAVEMENT LEGEND

- CONCRETE DRIVEWAY PAVEMENT
- CROSSOVER
- CONCRETE FOOTPATH PAVEMENT
- REINSTATEMENT TO COUNCIL REQUIREMENTS.
- PAVED DRIVEWAY
- PERMEABLE PATHWAY

WARNING
BEWARE OF UNDERGROUND SERVICES
THE CONTRACTOR WILL LOCATE ALL SERVICES ON SITE PRIOR TO COMMENCEMENT OF WORK.

MELBOURNE ONE CALL SYSTEM — DIAL 1100 — DIAL BEFORE YOU DIG

LAND MUST SLOPE TOWARDS WATER COLLECTION PITS.

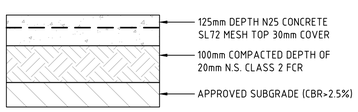
BATTER SOIL (IF APPLICABLE) AND INSTALL SPOON DRAIN AS REQUIRED. SPOON DRAIN TO BE CONNECTED TO PROPOSED STORM WATER SYSTEM. TYPICAL

RETAINING WALL (IF ANY) TO STRUCTURAL ENGINEER'S DETAILS. ALL AGI PIPES TO BE CONNECTED TO STORMWATER SYSTEM. TYPICAL.

WSUD NOTE:
CATCHMENTS AREA TO EACH RAINWATER TANK/RAINGARDEN TO BE AS PER APPROVED WSUD PLAN TO COUNCIL SATISFACTION. ANY DISCREPANCY MUST BE REFERRED BACK TO DESIGN ENGINEER PRIOR TO CONSTRUCTION. TYPICAL

NOTE:
- ALL PIT COVER LEVELS AND TOP OF I.O. TO BE ABOVE THE RL OF TOP OF BAFFLE WALL (IF APPLICABLE)
- MINIMUM 200mm GAP TO BE PROVIDED BETWEEN DETENTION PIPES.
- PLANTING OF TREES SHOULD BE AVOIDED NEAR THE PROPOSED PIPES OR PITS.
- THESE PLANS MUST BE READ IN CONJUNCTION WITH WSUD PLANS (IF APPLICABLE)

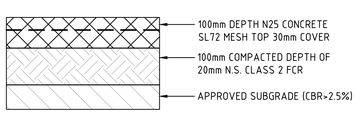
PROVIDE 2-N12 BARS × 1200 LONG TOP FOR ALL RE-ENTRY CORNERS OF PITS.



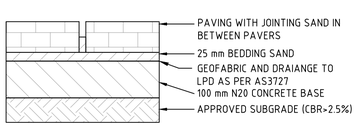
CONCRETE DRIVEWAY PAVEMENT
NTS



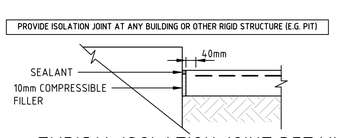
TYPICAL DRIVEWAY SECTION
NTS



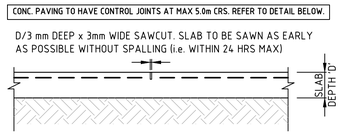
CONCRETE FOOTPATH PAVEMENT
NTS



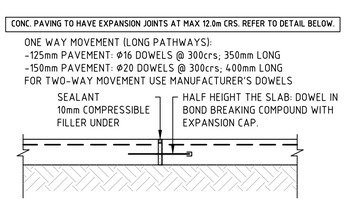
PAVED DRIVEWAY DETAIL
NTS



TYPICAL ISOLATION JOINT DETAIL
NTS



TYPICAL CONTROL JOINT DETAIL
NTS



TYPICAL EXPANSION JOINT DETAIL
NTS

Rev.	Remark/Comment	Date:	Appr.
	-		
	-		
A	CONSTRUCTION ISSUE	08.08.2025	D.M
P2	PRELIMINARY ISSUE	07.08.2025	D.M
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Client: MERAQ BUILDING DESIGNERS

Address: NO 189 MELBOURNE ROAD WILLIAMSTOWN

Designed: D.M.	Drawn: P.M
Drawing size: A1	Scale: AS NOTED
Job Number: 263135-C	Sheet No: 1 of 2
	Revision: A

GENERAL

G1. THESE DRAWINGS SHALL BE READ IN CONJUNCTION WITH ALL ARCHITECTURAL AND OTHER CONSULTANT'S DRAWINGS AND SPECIFICATIONS AND WITH SUCH OTHER WRITTEN INSTRUCTIONS AS MAY BE ISSUED DURING THE COURSE OF THE CONTRACT.
G2. ANY DISCREPANCY IN THE DOCUMENTATION SHALL BE REFERRED TO THE ENGINEER FOR WRITTEN INSTRUCTION PRIOR TO COMMENCING WORKS.
G3. ALL MATERIALS AND WORKMANSHIP SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF THE CURRENT AUSTRALIAN STANDARDS AND OF ALL RELEVANT AUTHORITIES, EXCEPT WHERE VARIED BY THE PROJECT DOCUMENTATION.
G4 ALL DIMENSIONS ARE TO BE OBTAINED FROM THE ARCHITECT'S DRAWINGS OR FROM SITE. ENGINEER'S DRAWINGS MUST NOT BE SCALED.
G5. DURING CONSTRUCTION THE CONTRACTOR WILL BE RESPONSIBLE FOR MAINTAINING THE STRUCTURE IN A STABLE CONDITION AND ENSURING NO PART OF THE STRUCTURE WILL BE OVERSTRESSED UNDER CONSTRUCTION ACTIVITIES. AS4801 AND THE REQUIREMENTS OF ALL RELEVANT AUTHORITIES ARE TO BE COMPLIED WITH.
G6. THE DESIGN/DRAWINGS ARE PART OF THE BUILDING CONTRACT ENTERED INTO BY THE CLIENT. ANY SUBSTITUTION WILL VOID THE CERTIFIED DESIGN. WRITTEN AUTHORITY FOR SUBSTITUTIONS CAN ONLY BE GIVEN BY THE SUPERINTENDENT AND ONLY THEN SUBJECT TO RE-CERTIFICATION.
G7. ANY DEFECT OR DEVIATION FROM THE CONTRACT DOCUMENTS THAT REQUIRES ASSESSMENT AND DETERMINATION OF CORRECTIVE MEASURES BY THE ENGINEER SHALL BE UNDERTAKEN AT THE CONTRACTOR'S EXPENSE.
G8. WATERPROOFING TO ARCHITECTS DETAILS.
G9. A MINIMUM OF 48 HOURS NOTICE SHALL BE GIVEN FOR ALL WORKS REQUIRING ATTENDANCE ON SITE BY THE ENGINEER. CONTRACTOR TO ENSURE THAT SUBJECT WORKS ARE COMPLETE PRIOR TO INSPECTION.
G10. DRAINAGE AROUND/UNDER STRUCTURES TO FOLLOW AS 2870 (SEE STRUCTURAL DESIGN).
G11. PROVIDE BLOCK ANCHORS TO STEEP PIPES AS PER STANDARDS.
G12. REFERENCE FOR USE OR INSTALLATION OF ANY MATERIALS ON THESE PLANS IS OUTSIDE THE SCOPE OF CERTIFICATION OF THE PLANS.

GENERAL CIVIL NOTES

GC1. FOR SITE SET OUT DIMENSIONS REFER TO ARCHITECTS DRAWINGS.
GC2. CONTRACTOR IS TO LOCATE ALL EXISTING SERVICES ON THE SITE AND IN SURROUNDING ROADS AND EASEMENTS PRIOR TO COMMENCEMENT OF EXCAVATION. ANY SERVICES DAMAGED BY THE CONTRACTOR TO BE REPAIRED AT FULL COST TO CONTRACTOR.
GC3. CONTRACTOR IS TO ARRANGE WITH THE RELEVANT STATUTORY AUTHORITIES FOR ALL NECESSARY INSPECTIONS AND APPROVALS. ALL WORK WITHIN ROAD RESERVES ETC. TO BE CARRIED OUT TO THE SATISFACTION OF THE RELEVANT AUTHORITIES.
GC4. ALL CONCRETE FOR KERB, KERB AND CHANNEL, PITS AND PEDESTRIAN PAVING TO HAVE A MINIMUM STRENGTH OF 32MPa AT 28 DAYS. ALL CONCRETE FOR VEHICULAR PAVING TO BE 32MPa. U.N.O.
GC5. SURPLUS EXCAVATED MATERIAL SHALL BE REMOVED FROM THE SITE.
GC6. ALL NEW WORKS SHALL MAKE A SMOOTH JUNCTION WITH EXISTING.
GC7. THE CONTRACTOR SHALL ARRANGE FOR THE WORKS TO BE SET OUT BY A LICENCED SURVEYOR.
GC8. THE CONTRACTOR SHALL INSTALL AND MAINTAIN SOIL EROSION AND SEDIMENT CONTROL MEASURES AS NECESSARY TO PREVENT RUN-OFF FROM THE SITE OF SEDIMENT RESULTING FROM HIS WORKS. SUCH MEASURES SHALL COMPLY WITH REQUIREMENTS OF COUNCIL.
GC9. ALL PAVEMENT SUBGRADE AREAS ARE TO BE STRIPPED OF ALL SIGNIFICANT ORGANIC SOILS TO A MINIMUM DEPTH OF 150mm. THE EXPOSED AREA IS TO BE PROOF ROLLED AND ANY SOFT SPOTS THAT ARE IDENTIFIED ARE TO BE REMOVED AND REPLACED WITH APPROVED GRANULAR MATERIAL COMPACTED IN 150MM LAYERS TO A DENSITY NO LESS THAN THAT OF THE SURROUNDING GROUND. PRIOR TO PLACING PAVEMENT SUB-BASE MATERIAL, THE UPPER LAYER OF SUBGRADE IS TO BE COMPACTED TO 100% STANDARD DENSITY AND TESTED IN ACCORDANCE WITH AS1289.

DRAINAGE NOTES

D1. STORMWATER DRAINAGE SHALL BE INSTALLED IN ACCORDANCE WITH AS3500.
D2. ALL STORMWATER PIPES TO BE APPROVED UPVC (SN4 OR SN6 FOR GENERAL DOMESTIC USE; SN8 OR SN10 FOR REDUCED COVER, UNDER PAVEMENTS & BUILDINGS AND WHEN SUBJECT TO BACKFILL/SURCHARGE) OR REINFORCED CONCRETE RUBBER RING JOINTED CLASS 2 TO CONFORM WITH AS4058 OR F.R.C. TO CONFORM WITH AS4139 OR HDPE PE100 CLASS SDR 13.9 MIN TO CONFORM WITH THE REQUIREMENTS OF AS4130 AS NOTED ON THE DRAWINGS. U.N.O.
D3. ALL OBSTRUCTING VEGETATION TO BE REMOVED BY CONTRACTOR.
D4. THE CONTRACTOR IS TO CONFIRM ALL SERVICE LOCATIONS PRIOR TO COMMENCING WORK ON UNDERGROUND SERVICES.
D5. ALL HD SEWER GRADE PIPE SHALL BE MINIMUM CLASS SDR13.6 COMPLYING TO AS4130.
D6. ALL DOWN PIPE CONNECTIONS TO BE 150 DIAMETER HD SEWER GRADE UNLESS NOTED OTHERWISE, LAID AT A MINIMUM GRADE OF 1 IN 100 WITH MINIMUM COVER OF 300mm. ANY DP CONNECTION BENEATH TRAFFICABLE PAVEMENT TO BE CAST IRON.
D7. ALL TRENCHES REQUIRED TO BE DEEPER THAN 1.5M SHALL CONFORM WITH THE APPLICABLE REQUIREMENTS OF THE OCCUPATIONAL HEALTH AND SAFETY ACT AND THE HEALTH AND SAFETY ORGANIZATION.
D8. CONTRACTOR TO MAINTENANCE CLEAN ALL NEW AND EXISTING STORMWATER DRAINAGE PRIOR TO LEAVING THE SITE.

SITE DRAINAGE IN REGARDS TO BUILDINGS

INADEQUATE SURFACE DRAINAGE SYSTEM IS A COMMON PROBLEM IN FOUNDATION MOVEMENT IN MODERATE/REACTIVE CLAYS. THE EFFECTIVE DRAINAGE SYSTEM THAT DOES NOT AFFECT FOUNDATION MOISTURE EQUILIBRIUM IS OF PARAMOUNT IMPORTANCE. BUILDER IS TO COMPLY WITH THESE REQUIREMENTS (OR NOTIFY THE CLIENT/OWNER IN CASE WORKS DO NOT FORM PART OF THE CONTRACT). THESE DRAINAGE REQUIREMENTS (NOT EXCLUDING ANY OTHER REQUIREMENTS SHOWN ON THESE PLANS) ARE TO BE ADHERED TO:

GENERAL DRAINAGE REQUIREMENTS:

1. SURFACE DRAINAGE OF THE SITE SHOULD BE CONTROLLED FROM THE START OF THE SITE PREPARATION AND CONSTRUCTION. DRAINAGE SYSTEM SHALL BE COMPLETED BY THE FINISH OF THE BUILDING CONSTRUCTION.
2. THE BASE OF TRENCHES SHALL BE SLOPED AWAY FROM THE BUILDING. TRENCHES SHALL BE BACKFILLED WITH CLAY IN THE TOP 300mm WITHIN 1.5m OF THE BUILDING. THE CLAY USED FOR BACKFILLING SHALL BE COMPACTED. WHERE PIPES PASS UNDER THE FOOTING SYSTEM THE TRENCH SHALL BE BACKFILLED FULL DEPTH WITH CLAY OR CONCRETE TO RESTRICT THE INGRESS OF WATER BENEATH THE FOOTING SYSTEM.
3. WHERE PIPES PASS UNDER THE FOOTING SYSTEM THE TRENCH SHALL BE BACKFILLED FULL DEPTH WITH CLAY OR CONCRETE TO RESTRICT THE INGRESS OF WATER BENEATH THE FOOTING SYSTEM. ALTERNATIVELY, A PLASTIC MEMBRANE ACROSS THE CROSS SECTION OF THE TRENCH, TAPED TO THE PIPE AND KEYED INTO THE SIDES AND BASE OF THE TRENCH MAY BE USED.
4. SUBSURFACE DRAINS TO REMOVE GROUNDWATER SHALL NOT BE USED WITHIN 1.0m OF THE BUILDING U.N.O.

GENERAL PLUMBING REQUIREMENTS:

1. PENETRATIONS THROUGH THE FOOTINGS SHOULD BE AVOIDED WHERE POSSIBLE.; OTHERWISE SHOULD BE CONSTRUCTED IN ACCORDANCE WITH DETAIL SON THESE PLANS.
2. CLOSED CELL POLYETHYLENE LAGGING SHALL BE USED AROUND ALL STORMWATER AND SANITARY PLUMBING DRAIN PIPE PENETRATIONS THROUGH FOOTINGS. THE LAGGING SHOULD BE MIN 40mm THICK FOR CLASS H2 AND E SITES AND CAN BE REDUCED TO 20mm FOR CLASS H1 SITES (SLEEVES ALLOWING EQUIVALENT MOVING CAN SE BE USED IN LIEU OF LAGGING).
3. DRAINS ATTACHED TO OR EMERGING FROM UNDERNEATH THE BUILDING SHALL INCORPORATE FLEXIBLE JOINTS IMMEDIATELY OUTSIDE THE FOOTING AND COMMENCING WITHIN 1.0m AWAY OF THE PERIMETER TO ACCOMMODATE DIFFERENTIAL SOIL MOVEMENT IN ANY DIRECTION EQUAL TO THE ys VALUE ON THE SOIL REPORT. THE FITTINGS OR DEVICES PROVIDED TO ALLOW FOR THE MOVEMENT SHALL BE SET AT THE MID POSITION OF THEIR RANGE OF POSSIBLE MOVEMENT AT THE TIME OF THEIR INSTALLATION, SO AS TO ALLOW FOR BOTH UPWARDS AND DOWNWARDS MOVEMENTS. THE REQUIREMENTS APPLIES TO ALL STORMWATER AND SANITARY PLUMBING DRAINS AND DISCHARGE PIPES.
4. ON SITE WASTEWATER TREATMENT UNITS AND ASSOCIATED LAND APPLICATION AREAS SHALL BE LOCATED TO MINIMIZE SOIL MOISTURE INCREASE WITHIN THE FOUNDATION.
5. DRAINAGE UNDER SLAB SHOULD BE AVOIDED WHERE PRACTICABLE.
6. WATER SERVICE PIPES SHALL BE SHALL COMPLY WITH THE RELEVANT REQUIREMENTS.
7. COLD, HEATED OR HOT WATER PIPES SHALL NOT BE INSTALLED UNDER THE SLAB UNLESS PIPES INSTALLED WITHIN A CONDUIT SO.IN CASE OF PIPE LEAKAGE IT WILL BE NOTICED ABOVE THE SLAB RATHER THAN LEAK UNONOTICED UNDER THE SLAB.
8. STORMWATER PIPES LAID UNDER THE FOOTINGS SHOULD HAVE MINIMUM CLEARANCE OF 25mm AND SHOULD BE GUARDED FROM MECHANICAL DAMAGE.

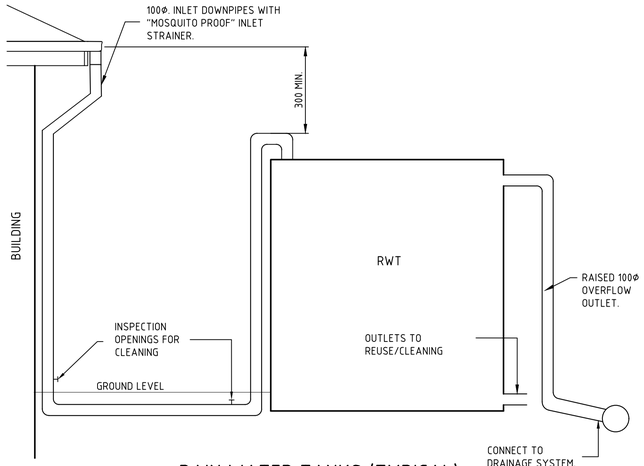
LANDSCAPING REQUIREMENTS:

1. THE GARDEN DEVELOPER SHALL NOT INTERFERE WITH THE DRAINAGE REQUIREMENTS, SUBFLOOR VENTILATION AND WEEPHOLE DRAINAGE SYSTEMS. GARDEN BEDS ADJACENT TO THE BUILDING SHALL BE AVOIDED. CARE SHOULD BE TAKEN TO AVOID OVER-WATERING OF GARDENS CLOSE TO THE FOOTING SYSTEMS.
2. PLANTING OF TREES SHOULD BE AVOIDED NEAR THE FOUNDATION OF THE BUILDING AR NEIGHBORING BUILDING AS THEY CAN CAUSE DAMAGE DUE TO DRYING EFFECTS. TO REDUCE (NOT ELIMINATE) THE POSSIBILITY OF DAMAGE TREES SHOULD RE RESTRICTED TO A DISTANCE FROM THE HOUSE:
- ONE AND A HALF MATURE HEIGHT FOR CLASS E SITES
- ONE MATURE HEIGHT FOR CLASS H1 AND H2 SITES
- THREE QUARTERS OF MATURE HEIGHT FOR CLASS M SITES.

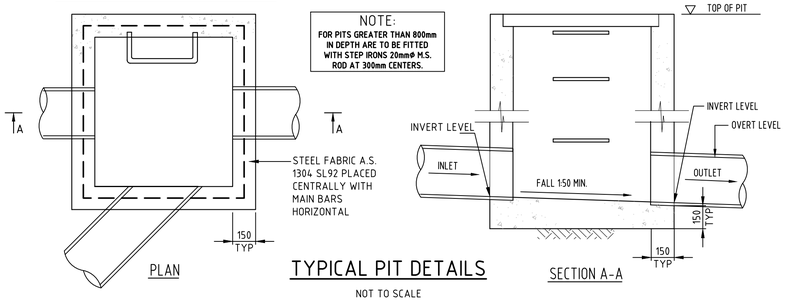
WHERE ROWS OR GROUP OF TREES ARE INVOLVED THE DISTANCE FROM THE BUILDING SHOULD BE INCREASED. REMOVAL OF TREES CAUSES SIMILAR PROBLEMS.

MAINTENANCE:

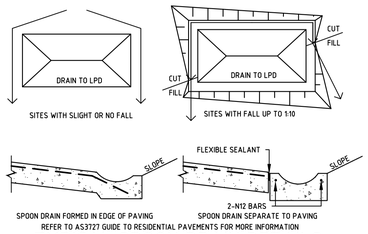
1. PRIMARY OBJECTIVE OF THE MAINTENANCE IS TO MINIMIZE SOIL MOISTURE CHANGES CLOSE TO/UNDER THE FOUNDATION.
2. LEAKS IN PLUMBING (INCLUDING STORMWATER AND SEWERAGE DRAINAGE) AND TAPS SHALL BE REPAIRED PROMPTLY. THE OWNER IS RESPONSIBLE FOR ROUTING INSPECTIONS OF THE DRAINAGE SYSTEM. SILT TRAPS SHOULD BE INSPECTED MINIMUM ANNUALLY AND BUILD UP IS TO BE CLEARED. HEAVY BUILD UP MAY REQUIRE FLUSHING OF THE AGGI DRAINS.



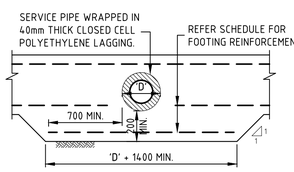
RAIN WATER TANKS (TYPICAL)
NOT TO SCALE



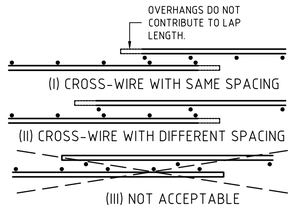
TYPICAL PIT DETAILS
NOT TO SCALE



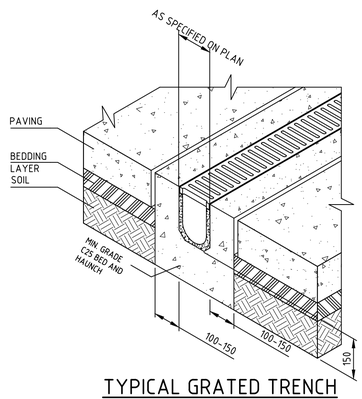
TYPICAL SPOON DRAIN DETAILS
SCALE NTS



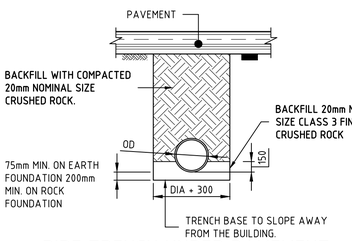
TYPICAL PENETRATION THROUGH FOOTING DETAIL
SCALE = NTS



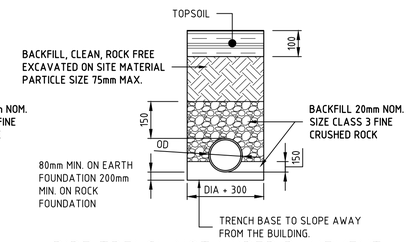
LAPS IN TOP MESH REINFORCEMENT
SCALE= NTS



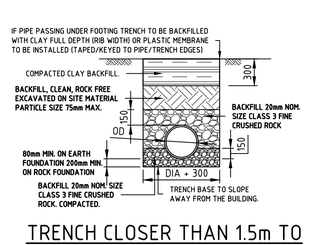
TYPICAL GRATED TRENCH
NOT TO SCALE



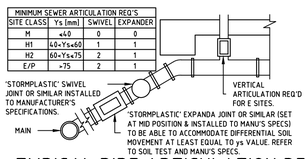
PIPE TRENCH UNDER PAVEMENT
NOT TO SCALE



PIPE TRENCH NOT UNDER PAVEMENT
NOT TO SCALE

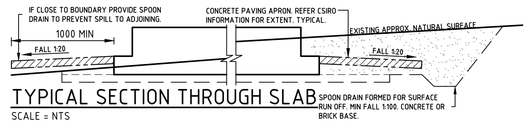


TRENCH CLOSER THAN 1.5m TO BUILDING BACKFILL DETAIL
NOT TO SCALE



MINIMUM SEWER ARTICULATION REQ'S			
SITE CLASS	Ys (mm)	SWIVEL	EXPANDER
M	<40	0	0
H1	40<Ys<60	1	1
H2	60<Ys<75	2	1
E/P	>75	2	1

TYPICAL PIPE ARTICULATION DETAIL
SCALE = NTS



TYPICAL SECTION THROUGH SLAB
SCALE = NTS

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A	CONSTRUCTION ISSUE	08.08.2025	D.M
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Client: MERAQ BUILDING DESIGNERS

Address: NO 189 MELBOURNE ROAD WILLIAMSTOWN

Designed: D.M.	Drawn: P.M
Drawing size: A1	Scale: AS NOTED
Job Number: 263135-C	Sheet No: 2 of 2
	Revision: A